

Network Performance

Bandwidth

→ Max rate / capacity at which a link can transmit data

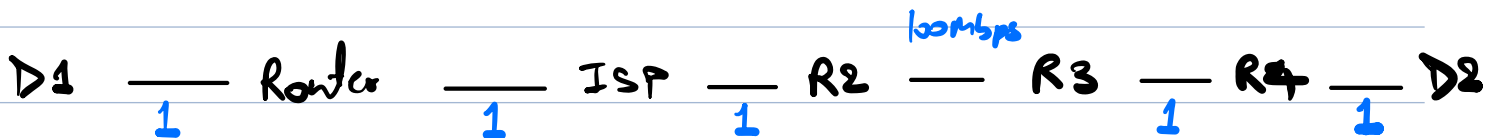
Kbps

Mbps

Gbps

Throughput

⇒ Actual rate at which data is successfully transferred.



Latency

Time taken for data to travel through a network

RTT (Round Trip time)

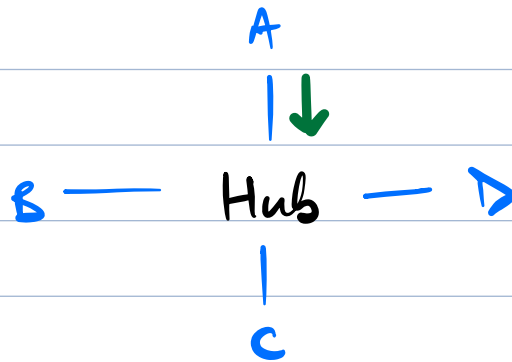
Network Devices

Router

Switch

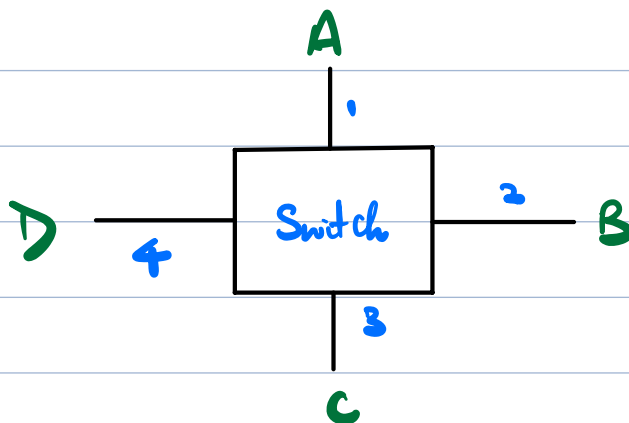
Hub

Hub



Switch

Devices connected within a local network & intelligently forwards Ethernet frames.



MAC Address

Routers

Networking Layers

laptop → Wifi → ISP → R1 → Internet → R2 → GS
(Browser)

Modularity.

Abstraction

Independent Evolution.

OSI Model (Open System Interconnection)

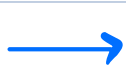
- 7) Application
- 6) Presentation
- 5) Session
- 4) Transport
- 3) Network
- 2) Data link
- 1) Physical

Application — HTTP, DNS
Transport — TCP, UDP
Network — IP
Data link — Ethernet, Wi-fi
Physical — Radio, Copper, Fiber.

Layer	Think about
Application	What does the application want?
Transport	Which process + how should it communicate?
Network	Where is the destination across networks?
Data Link	How do I reach the next/local device?
Physical	How are the bits transmitted?

Data Representation

"Hello"



ASCII values

72 → 01001000

Bit v/s Byte

Bit $\Rightarrow$ Single binary digit

Byte $\Rightarrow$ 8 bits

8 bits $\Rightarrow$ 1 Byte

1000 bits $\Rightarrow$ 1 Kilobit (kb)

1000 Bytes $\Rightarrow$ 1 KiloByte (KB)

1000 kb $\Rightarrow$ 1 Mb

1000 KB $\Rightarrow$ 1 MB

MB/s

Mbps

$$100 \text{ Mbps} \rightarrow \frac{100}{8} = 12.5 \text{ MB/s}$$

Transmission

$\Rightarrow$ How long does the sender takes to put all packet bits onto link ??

$$d_{\text{trans}} = \frac{L}{R} \rightarrow \begin{array}{l} \text{length of packet} \\ \text{Transmission rate} \end{array}$$

Propagation

How long does the physical signal takes to travel across the link

$$d_{\text{prop}} = \frac{\text{distance}}{\text{prop speed.}}$$

Bits



frames



contains IP packet



contains Transport data



contains Application Data

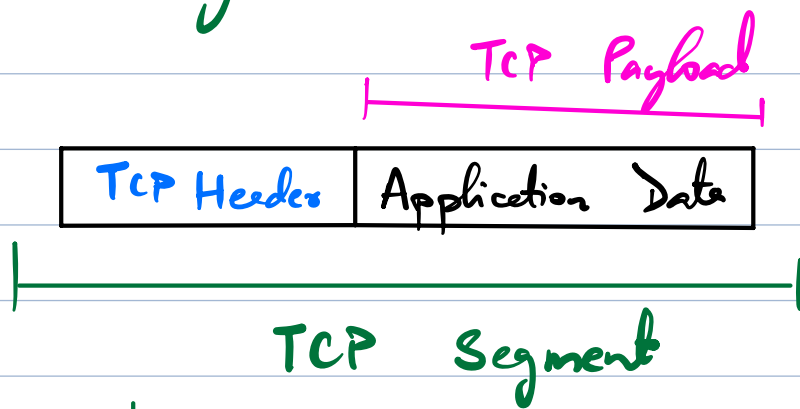
Application

GET /index.html HTTP/1.1

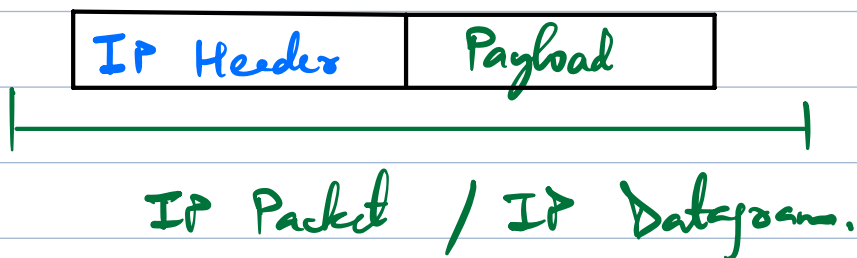
Application Data



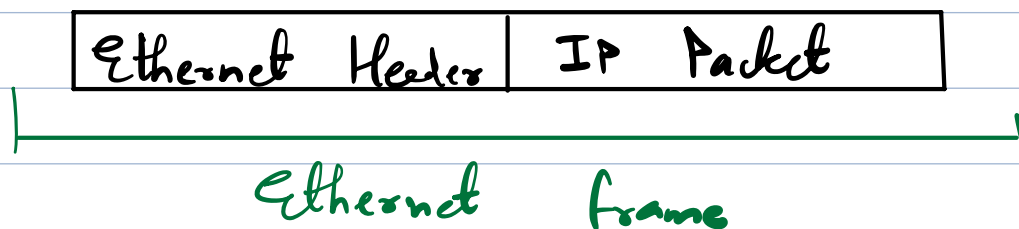
Transport Layer



Network Layer



Data link layer



Physical layer

[Ethernet [IP [TCP [Data]]] fcs]
Segment

Packet

Frame



Physical

Encapsulation is the process in which each networking layer wraps data from the layer above with the control information required by its protocol.

! What is a Payload.

One layer's complete protocol data unit generally becomes data/payload carried by the layer below.

Protocol Data Units (PDUs)

⇒ Generic term for the structured unit handled by a networking layer

Decapsulation

Receive Bits



Data link

~~[Ethernet [IP [TCP [Data]]]] fcs]~~



Network

~~[IP [TCP [Data]]]~~



Transport

~~[TCP [Data]]~~



Application : Receiver HELLO

Sender

Receiver

Application <-----> Application

Transport <-----> Transport

Network <-----> Network

Data Link <-----> Data Link

D1

Router

D2

A



T



N



L



P



Network



link



Physical



A



T



Net.



link



P

1) Application Data

GET /index.html HTTP 1.1
Host : www.google.co.in

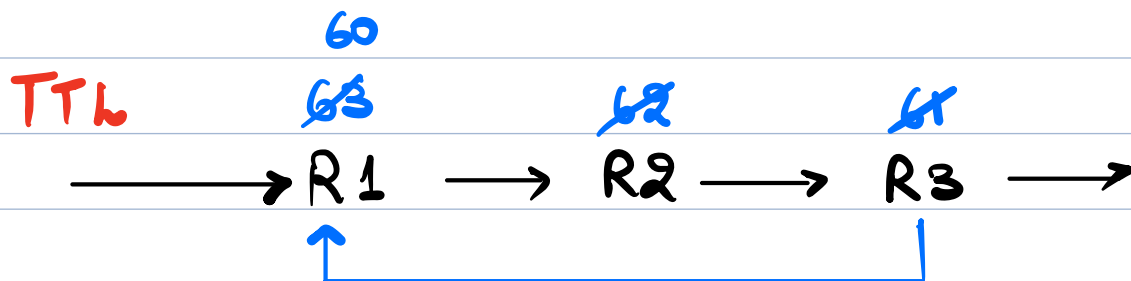
GET /index.html

2) TCP Header

Source Port	Dest Port	Seq / Ack	Data
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3) IP Layer

Source IP	Dest IP	Protocol	TCP Segment
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4) Ethernet Header

Dest. MAC	Source MAC	EtherType	IP Packet
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IP₁

IP₂

D1

R1

R2

D2

M1

M2

M3

M4

IP IP₁ → IP₂

IP IP₁ → IP₂

IP IP₁ → IP₂

IP IP₁ → IP₂

Eth M1 → M2

M2 → M3

M3 → M4

FCS → Frame Check Sequence

ETHERNET FRAME

Destination MAC
Source MAC

IP PACKET

Source IP
Destination IP
TTL

TCP SEGMENT

Source Port
Destination Port
Sequence / ACK

APPLICATION DATA
HTTP Request

FCS

MAC Address

IP Address

Port Number
